# Industrial Asset Maintenance Handbook

General Guidelines for Pumps, Chillers, Motors & HVAC Systems
Version 1.0 | Revision Date: 2024-06-01

## 1. General Safety Guidelines

**Electrical Safety**
• Always de-energize equipment before performing maintenance
• Use lockout/tagout procedures per OSHA requirements
• Test for absence of voltage using calibrated meter
• Wait 5 minutes after shutdown before opening enclosures
• Ground anti-static equipment when working with electronics

**Mechanical Safety**
• Use appropriate PPE: hard hat, safety glasses, work gloves, steel-toe boots
• Never wear loose clothing or jewelry near rotating equipment
• Assume all fluids are hot until proven otherwise
• Use proper lifting techniques for items >25 kg (55 lbs)
• Have spill kits available when servicing hydraulic/coolant systems

**Pressure System Safety**
• Release pressure gradually using proper bleed-down procedures
• Never strike or heat pressurized vessels or lines
• Isolate from power and use mechanical locks on isolation valves
• Verify pressure gauge reads zero before disconnecting
• Replace pressure relief valves annually or per manufacturer

## 2. Predictive Maintenance Techniques

**Vibration Analysis**
Vibration monitoring detects early failures in rotating equipment.
**ISO 20816 Standards:**
• Zone A (0-7.1 mm/s): Acceptable - new equipment baseline
• Zone B (7.1-11.2 mm/s): Just Acceptable - monitor closely
• Zone C (11.2-18 mm/s): Unacceptable - schedule maintenance within 1 week
• Zone D (>18 mm/s): Machine In Imminent Failure - shutdown immediately

**Thermal Imaging**
Detect hot spots indicating:
• Electrical connections: Loose connections create heat
• Motor bearings: High friction = high temperature
• Heat exchangers: Blockage or fouling reduces thermal transfer
• Hydraulic lines: Temperature rise indicates pressure drop or leakage

**Oil Analysis**
Monthly oil sampling reveals:
• Particle count: >1000 particles indicates wear

• Water content: >500 ppm suggests seal failure or ambient moisture
• Viscosity change: >±5% from baseline indicates degradation
• Acid number (TAN): >2 mg KOH/g suggests oxidation

**Acoustic Monitoring**
High-frequency sound analysis detects bearing degradation and cavitation:
• Normal bearings: 500-2000 Hz baseline sound signature
• Early wear: 2000-5000 Hz increase in harmonic content
• Cavitation: Distinctive crackling or popping sounds
• Frequency trending: Increasing trend indicates imminent failure

## 3. Common Failure Modes by Equipment Type

**Pump Failures**
1. Cavitation: Low suction pressure → vapor bubbles → erosion
Prevention: Maintain minimum NPSH, warm liquid carefully on startup
2. Seal failure: Chemical attack, heat degradation, pressure spikes
Prevention: Monitor temperature <45°C, control pressure ramps
3. Bearing wear: Inadequate lubrication, contamination, misalignment
Prevention: Oil change every 2000 hours, maintain alignment ±0.1mm

**Chiller Failures**
1. Refrigerant leakage: Corrosion, vibration fatigue, connection looseness
Prevention: Pressure test annually, monitor superheat
2. Heat exchanger fouling: Water scale, biological growth, corrosion
Prevention: Water treatment program, chemical cleaning annually
3. Compressor failure: Liquid slugging, oil starvation, overheating
Prevention: Maintain proper oil level, superheat >4°C

**Motor Failures**
1. Winding insulation breakdown: Thermal stress, moisture, contamination
Prevention: Keep ambient <50°C, enclosure IP54 or better
2. Bearing seals failure: Water ingress, contaminant infiltration
Prevention: Monitor leakage at drain holes, replace seals every 3 years
3. Rotor bar cracking: Thermal cycling, magnetic stress
Prevention: Avoid frequent start/stop cycles, ensure proper cooling

## 4. Recommended Inspection Schedule

| Equipment | Daily | Weekly | Monthly | Annually |
|---|---|---|---|---|
| Pump | Temp/Sound | Log readings | Vibration | Seal inspection |
| Chiller | Pressure/Temp | Oil level | Heat exchanger | Diagnostics |
| Motor | Temp/Amperage | Lubrication | Bearing play | Insulation test |
| HVAC | Filter | Drainage | Coil clean | System pressure |

## 5. Record Keeping & Documentation

Detailed maintenance records are essential for:
- Trending equipment performance over time
- Identifying patterns before catastrophic failures
- Supporting warranty claims
- Compliance with regulatory requirements
- Justifying capital equipment replacement

**Required Documentation:**
- Work order with date, time, technician name, nature of work
- Before/after sensor readings (temperature, pressure, vibration)
- Parts replaced with manufacturer part numbers and quantities
- Special observations, recommendations, safety issues
- Estimated time to complete and actual time spent